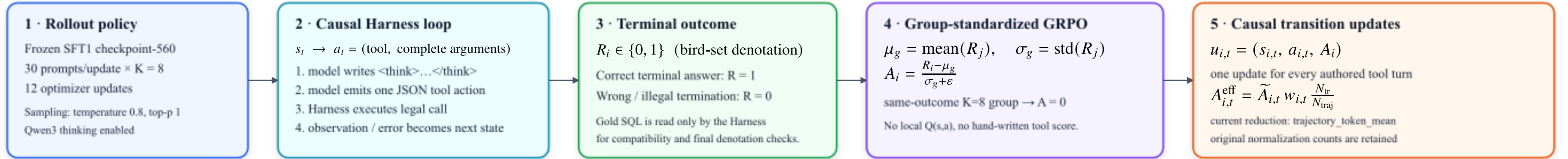


SAAM-Asymmetric GRPO for Atomic Tool-Use RL

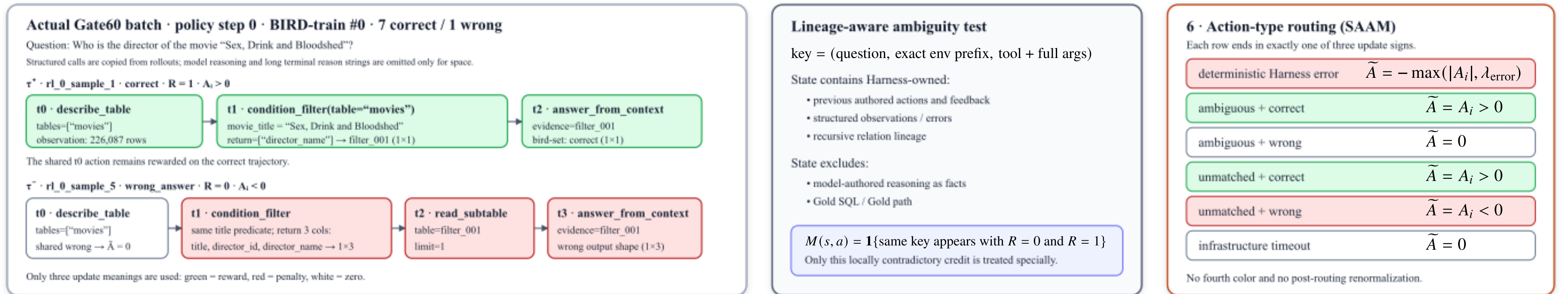
Qwen3-8B · Atomic version26 · result-only terminal supervision · train-only BIRD-train · Gold SQL never enters the actor

Research claim under test
remove contradictory credit, keep vanilla elsewhere

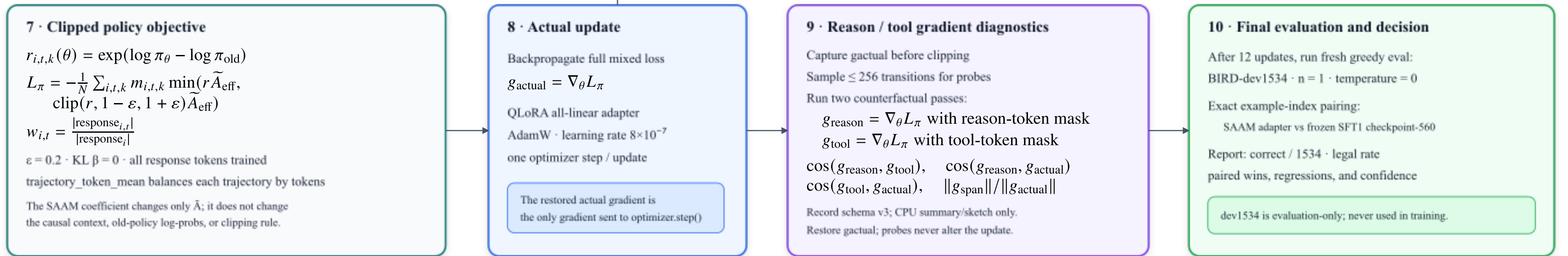
A. Causal rollout and terminal GRPO signal



B. What SAAM sees inside one question group



C. Tokenwise clipped update, diagnostics, and evaluation



Legend ■ reward: $\tilde{A} > 0$ ■ penalty: $\tilde{A} < 0$ ■ zero update: $\tilde{A} = 0$

All text: Times New Roman · equations: LaTeX + newtxmath
restore actual gradient → optimizer.step() → next rollout/update